

Computer Architecture

(History)

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Contents

- History of computer developments
- Modern computer systems

History of computer developments (~1980s)

Computer?

- The term “Computer” was originated from <people who calculate>
 - In the 18th century, people on ships calculated navigation routes
 - During the first World War in the 1910s, people calculated the target location of cannon, missile etc.
 - Young women were employed for wiring the ENIAC machine

Summary of developments

년도	발명기종	설명	방식
1642	계산기	프랑스 파스칼, 수동기계식 계산기 (가감산)	수동기계식
1671	계산기	독일 라이프니츠, 수동기계식 계산기 (사칙연산)	수동기계식
1822	미분엔진	영국 배비지, 최초의 자동기계식 계산기 설계, 나중에 일부 완성됨 (수학계산용)	자동기계식
1837	해석엔진	영국 배비지, 최초의 자동기계식 컴퓨터 설계, 설계 원리가 MARK I에서 실현됨 (범용)	자동기계식
1936	Z1	독일 쑤저, 최초의 전기기계식 컴퓨터 제작, 특허 출원, 최초의 2진 연산방식 (범용)	전기기계식
1937	ABC	미국 아이오와 주립대학 아타나소프 교수와 베리, 최초의 진공관식 계산기, 2진 (수학계산용)	진공관
1944	MARK I	미국 하버드대학 에이킨이 IBM과 공동개발, 최초의 전기기계식 컴퓨터로 알려짐, 10진	전기기계식

Summary of developments – Cont'd

1944	Colossus	영국 토미 플라워즈, 2진 연산방식 (암호해독용)	진공관
1946	ENIAC	미국 펜실베니아대학 에커트와 모클리, 최초의 전자식 컴퓨터 (범용), 외부 프로그램 방식, 10진	진공관
1949	EDSAC	영국 모리스 윌키스, 폰 노이만의 참여로 최초로 프로그램 내장 방식 채택, 2진 연산방식	진공관
1950	EDVAC	미국 에커트와 모클리, 폰 노이만 참여로 에니악을 개량해 프로그램 내장 방식 완성, 2진 연산방식	진공관
1951	UNIVAC I	에니악을 만든 미국 에커트와 모클리, 최초의 상업용 전자식 컴퓨터, 미국 통계국에서 구입	진공관
1953	IBM 701	미국 IBM의 상업용 컴퓨터 판매 시작	진공관
1964	IBM s/360	IBM의 system/360, 370은 한 때 메인프레임의 대명사	IC
1970	IBM s/370	대규모집적회로(LSI)로 만든 전자식 컴퓨터	LSI
1971	intel 4004	최초의 마이크로프로세서	
1976	CRAY-I	최초의 상업용 슈퍼컴퓨터	

Blaise Pascal (1623~1662)

- Invented the first mechanical calculator
 - Used the concept of **saw-toothed wheel**
 - Machine for **decimal addition** and **subtraction** (+, -)



Year
1642



Gottfried Leibniz (1646~1716)

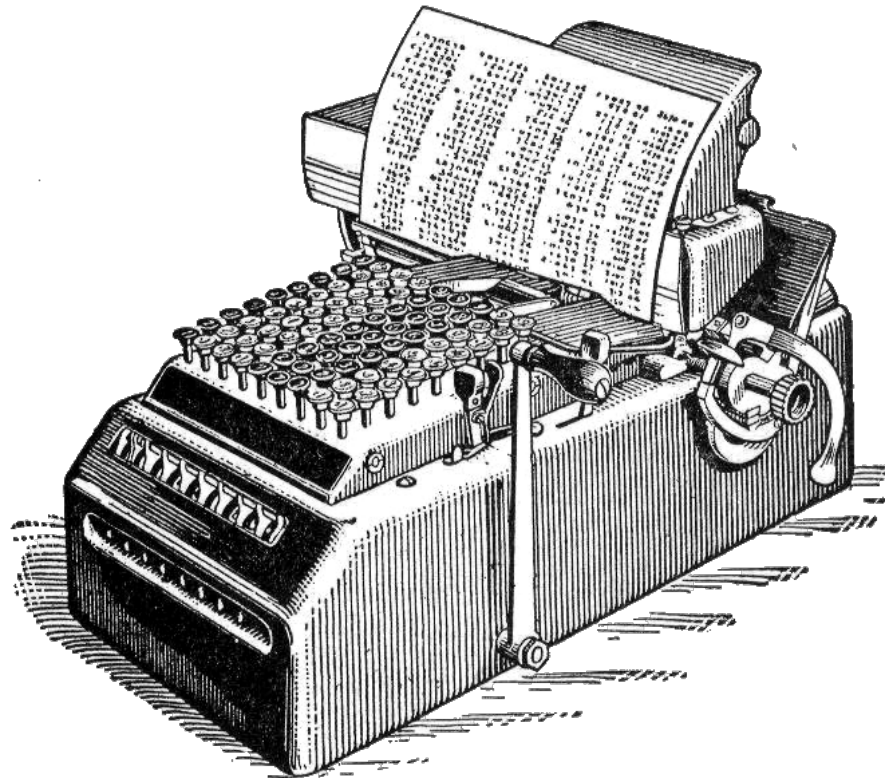
- Invented the first calculator of the **four fundamental arithmetic operations** (+, -, $\times$, $\div$)
 - Introduced **gears** into Pascal's calculator
 - Multiplication by repeating additions
 - Division by repeating subtractions



Year
1671

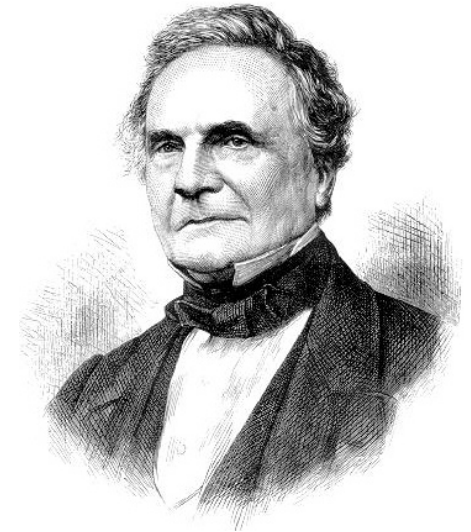
Mechanical computer

- Till the mid of 20th century
 - Manual mechanical calculator



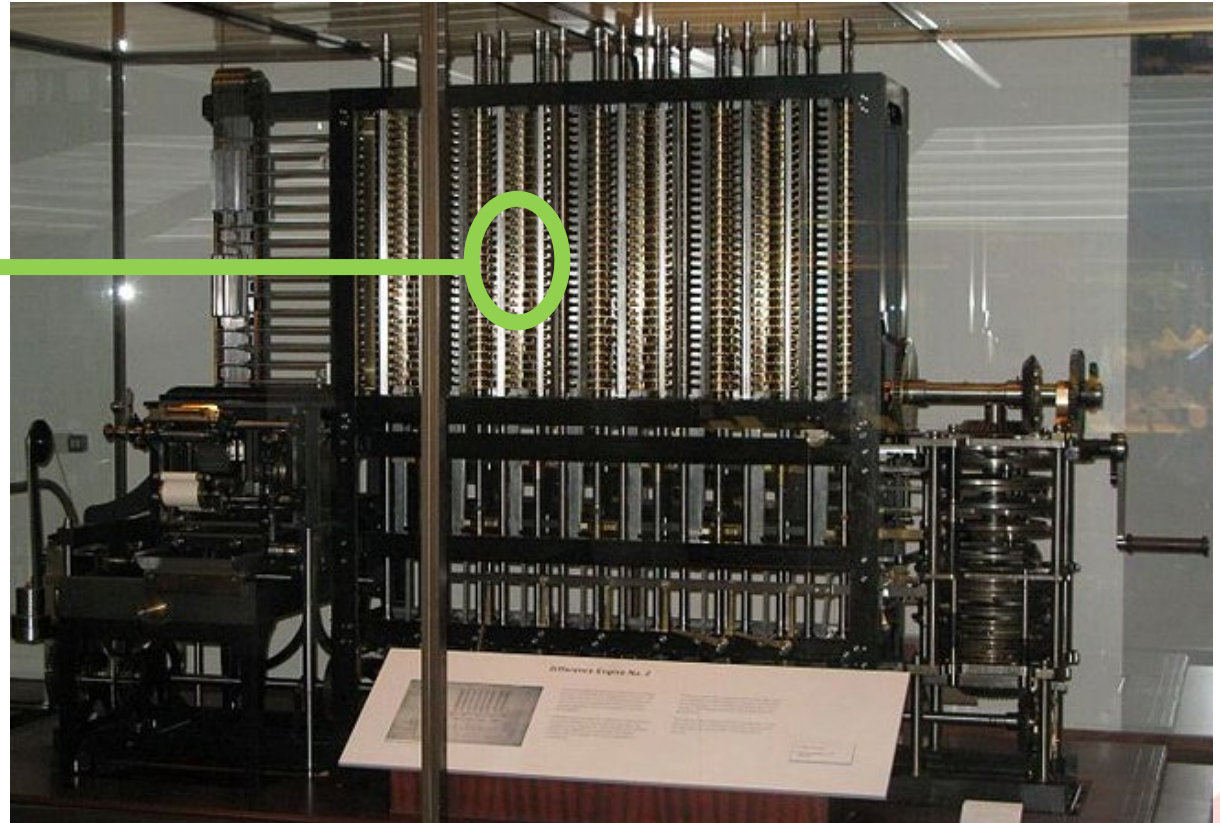
Charles Babbage (1791~1871)

- Invented a machine which can calculate all kinds of calculations
- Designed the first **difference engine** (1822)
 - To calculate $\log()$ and trigonometrical functions
 - **Automatic mechanical** calculator
 - At that time, it wasn't success as all the parts were designed mechanically



Charles Babbage (1791~1871) – Cont'd

- Difference engine, 1822
 - The Science Museum (London)

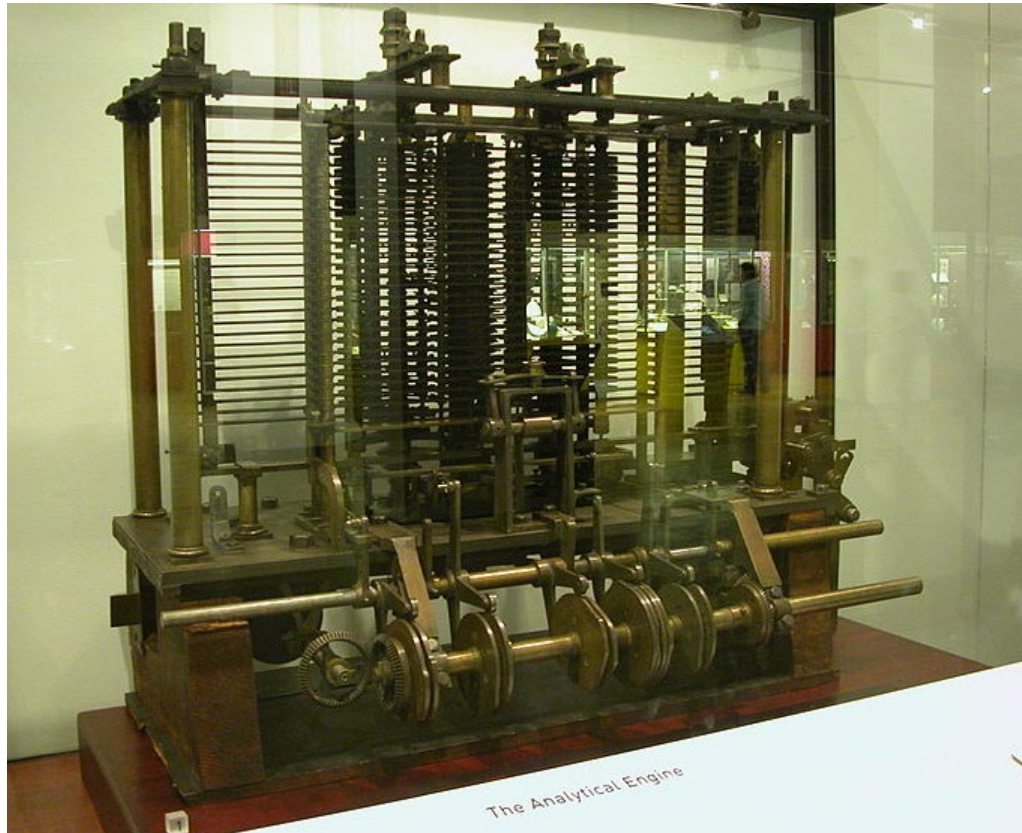


Charles Babbage (1791~1871) – Cont'd

- Designed the first **analytical engine** (1837)
 - The **original design of modern computers**
 - Input/output devices, storage unit, processing unit, control unit
 - The **punched card** reader decoded the instructions of the calculation cards
 - The computing device **automatically** performed the calculations
 - The results were then printed using the card punch printer
 - Intended for **general-purpose programming**
 - All components were designed **mechanically** and **powered by steam engine**
 - Was not successful at that time, but eventually realized by MARK I (1944)

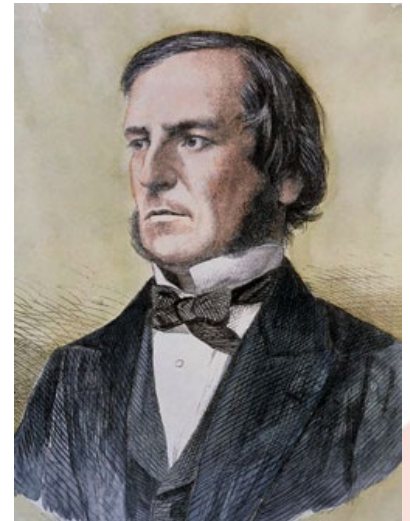
Charles Babbage (1791~1871) – Cont'd

- Analytical engine, 1837
 - The Science Museum (London)



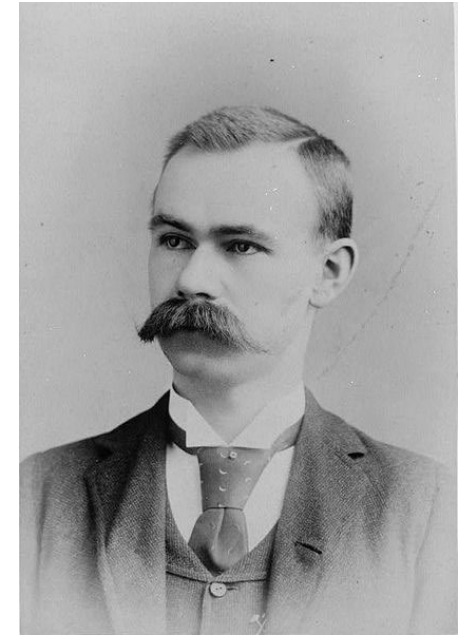
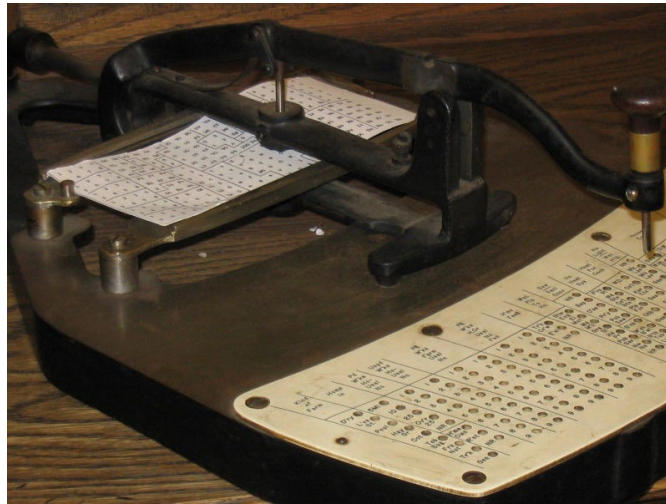
Ada Lovelace and George Boole

- Countess Ada Lovelace (1815~1852)
 - Assisted C. Babbage in designing the program for his analytical engine
 - She is often referred to as the world's first programmer
- George Boole (1815~1864)
 - The founder of modern symbolic logic
 - Established mathematical theories of logic and probability
 - Invented Boolean algebra, the representative form of mathematical logic
 - Logical operations such as AND, OR, NOT etc.



Herman Hollerith (1860~1929)

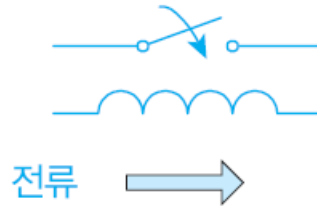
- Invented the first **statistical processing machine**, punch card system
 - Revolutionized the 1890 US population census
 - Reduced the processing time **from years to weeks**
 - Demonstrated the effectiveness of **batch processing of data**



Founded a
company that
become IBM

Eletromechanical computer

- Combining **electrical** and **mechanical** components
- Used the operating principle of **relay**
 - When current flows, the contacts stick to the electromagnet **mechanically**
 - Representing **binary numbers** through mechanical switching
 - Like a semi-conductor, transmits 1s and 0s in digital logic circuits



Konrad Zuse (1910~1995)

- The first known **binary digital computer model**, namely Z1
- Later model Z3 was capable of performing **floating-point arithmetic** operations
 - It is classified as the **first programmable computer**
 - Data processing was divided into input devices, memory, registers, arithmetic units, output devices, and programming language research
 - It **stored programs in memory before** the von Neumann architecture



Konrad Zuse (1910~1995) – Cont'd

- Z1 (1936)



Berlin German
Museum of
Technology

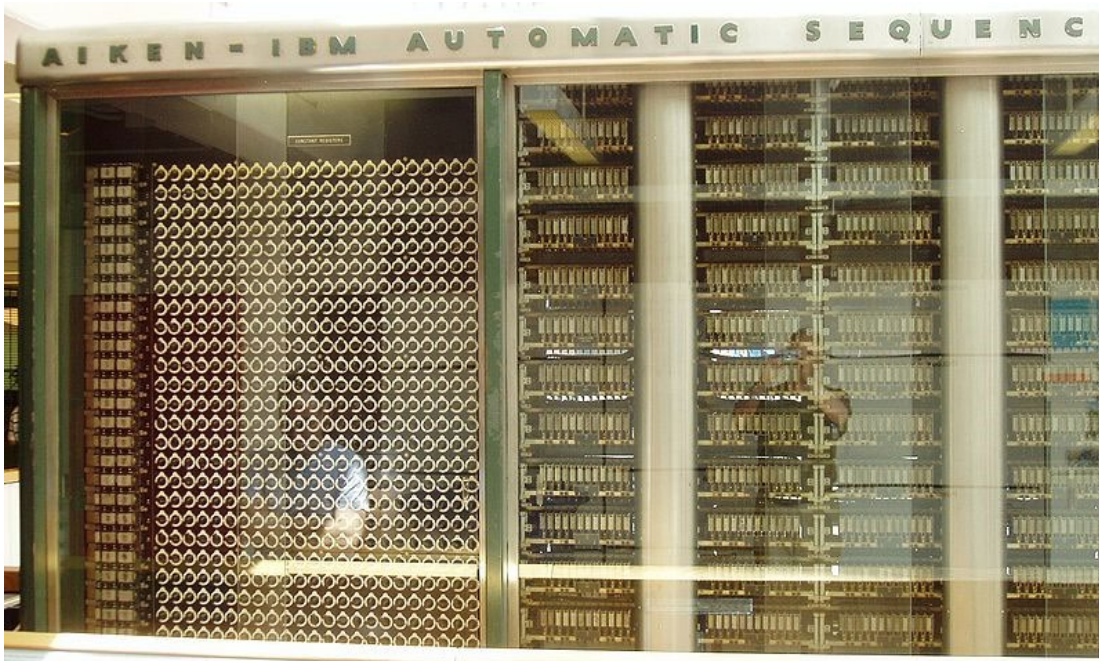
- Z3 (1941)

Deutsches Museum
in Munich



Howard H. Aiken (1900~1973)

- Since 1937 with support from IBM, the design of a **general-purpose electro-mechanical computer** was undertaken, resulting in the completion of the Harvard **Mark I** in 1944



MARK I (1944) computer ➡ Harvard University

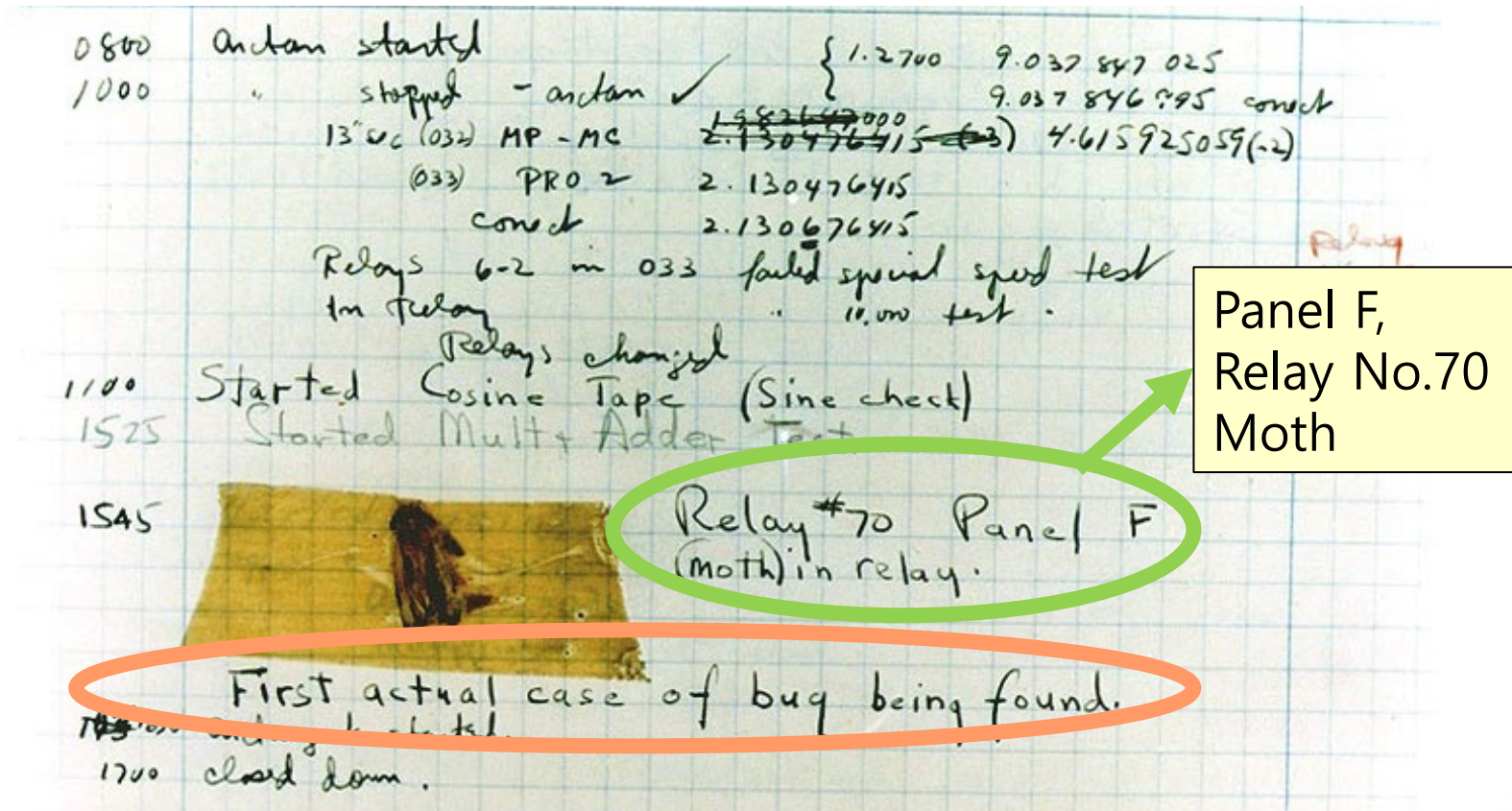
Grace M. Hopper (1906~1992)

- The first female admiral in the history of the US Navy
- Developed the **world's first compiler**
 - Primitive interpreter
- Mother of COBOL language
- Firstly, used the term **“bug”** in computer history
 - The reason for the malfunction of the MARK II in 1947 was that a **moth** got stuck in the relay



Grace M. Hopper (1906~1992) – Cont'd

- First actual case of bug being found



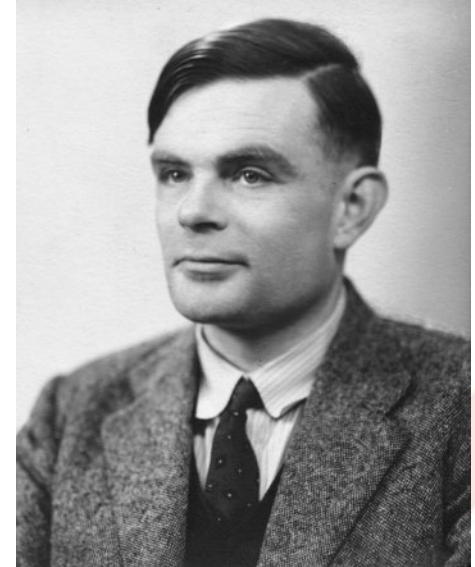
👉 Smithsonian Museum, USA

Bug and debugging

- Bug in programming
 - Errors or faults in programs that cause inaccurate results or crashes
 - Hardware malfunction issue
- Debug or debugging
 - Bug removal
 - The process of finding and correcting coding errors in a program

Alan Turing (1912~1954)

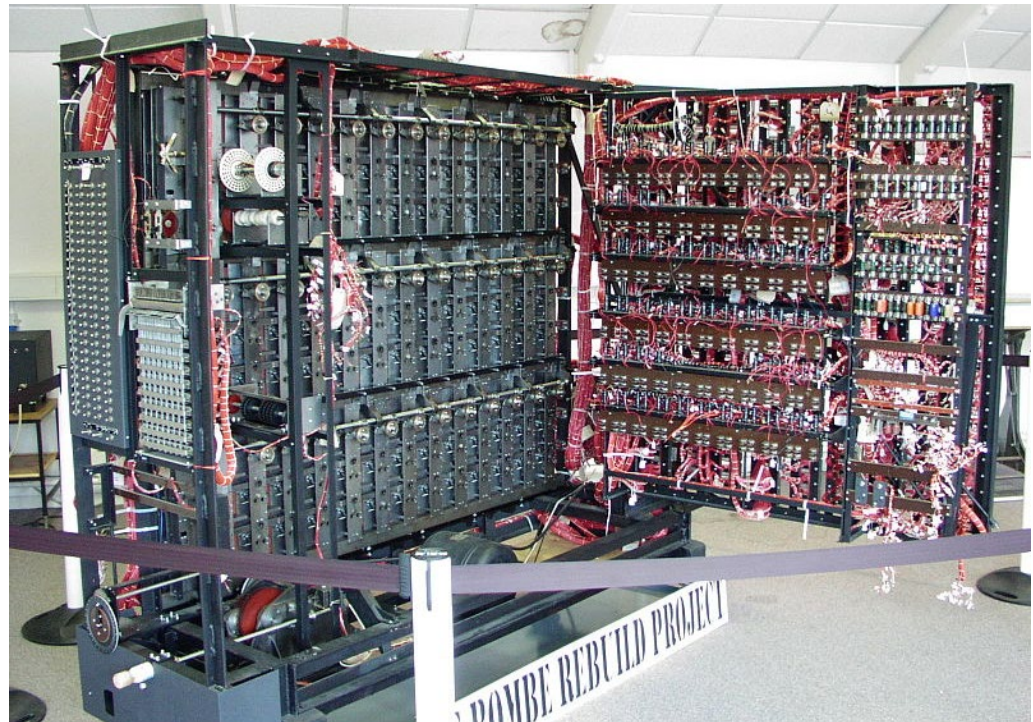
- A mathematician from UK
- <On Computable Numbers, with an Application to the Entscheidungs problem, 1936>
 - Master's thesis, the first study on how far machines can operate logically
- Turing machine
 - Infinite storage capacity and a faultless kind of virtual machine
 - A **mathematical model** for modern computers today
- A movie about Alan Turing
 - The imitation game, 2014
 - About enigma



Alan Turing (1912~1954) – Cont'd

- Turing's bombe (1940)
 - At the time of World War II, German army used encryption machine **Enigma**
 - Turing's bombe was used to decrypt

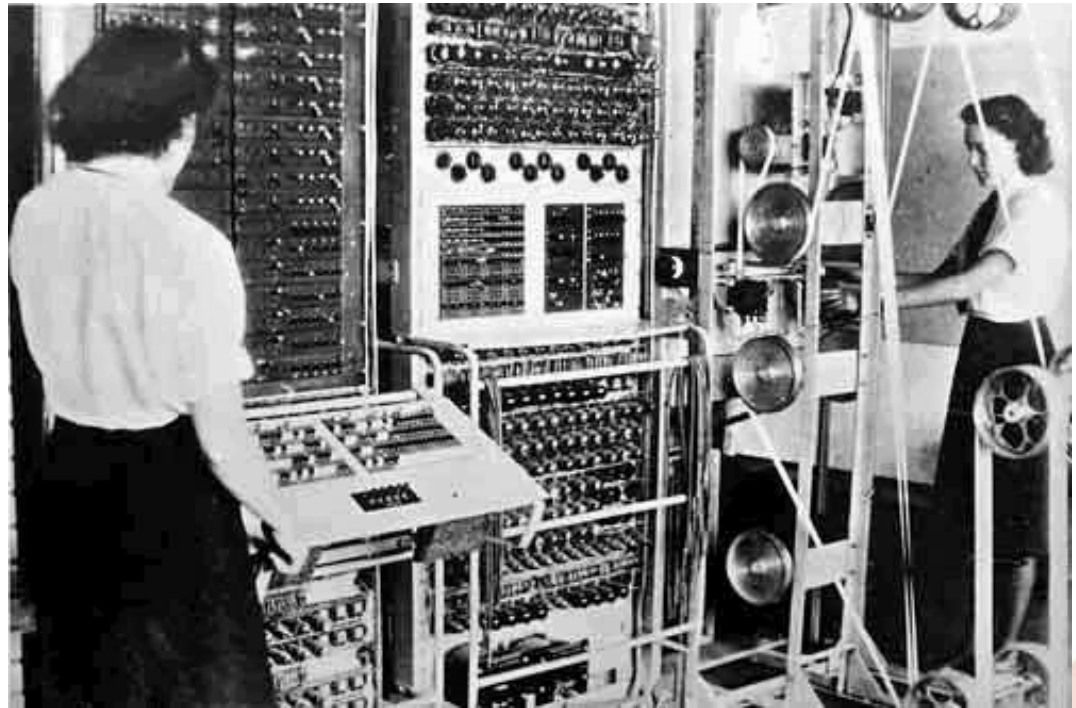
Bletchley Park
Museum, UK



Tommy Flowers (1905~1998)

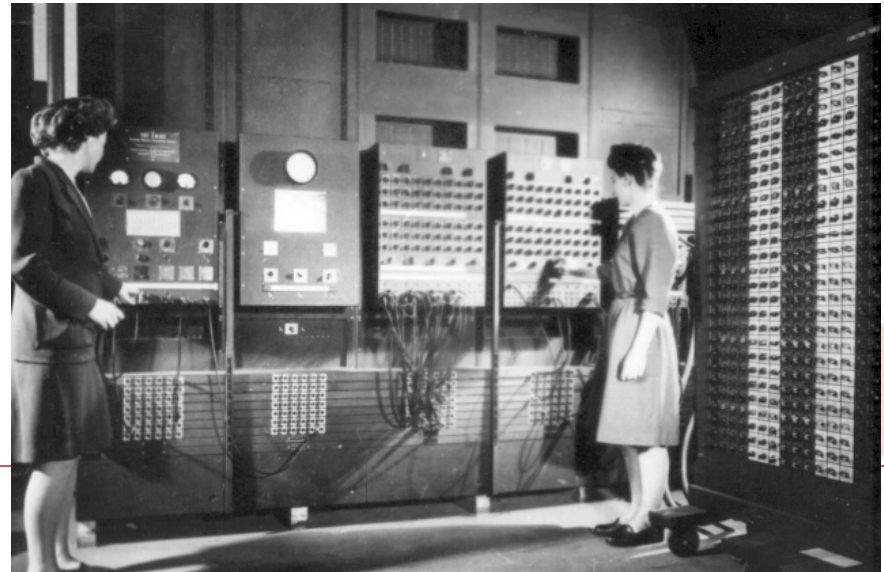
- Designed **Colossus computer** (1944)
 - The world's first **electronically** programmable computer

Vacuum tubes
and relay



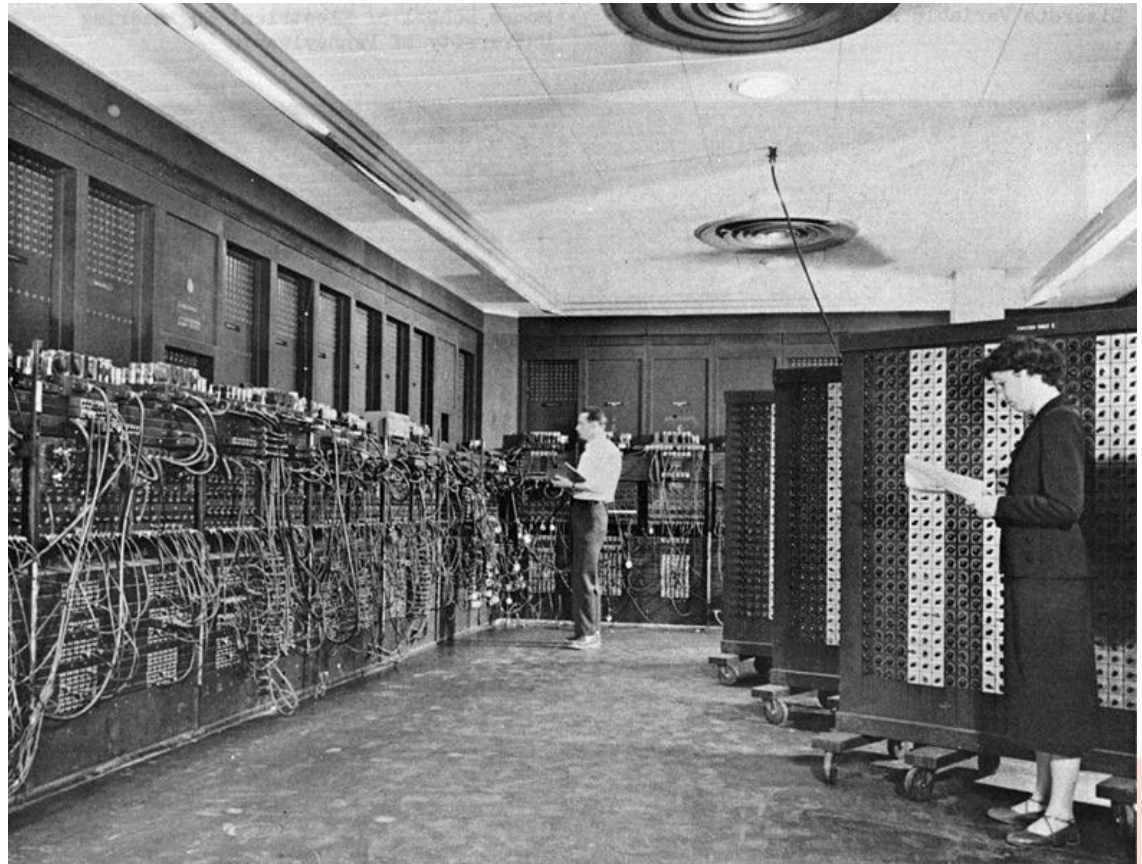
ENIAC Computer

- Electronic numerical integrator and calculator (ENIAC, 1946)
 - Invented by J. Presper Eckert and John Mauchly of UPenn
 - The **world's first general-purpose electronic computer** made with **vacuum tubes**
 - Designed as a machine to calculate **ballistic trajectories** for **weapon firing** with supports from the US Department of Defense
 - A computing device **adjusted by hardware** instead of software
 - Primitive job-based computer programming
 - An inconvenient **external programming** method using electric wiring, switches, plugboards, and punch cards for each operation



ENIAC Computer – Cont'd

- Size (H×D×W) : 2.4m×0.9m×30m
 - 167m²
 - Weight: 27t
 - Power: 150kW



Limitations of ENIAC

- Takes **a few days** in programming
 - Relied on thousands of switches and plugboards without any memory
 - A possible resolution was to pre-store programs within the **computer's internal memory**



Maurice Wilkes (1913~2010)

- Invented **Electronic delay storage automatic calculator (EDSAC)** in 1949
 - The world's first computer with **program stored internally**
- Firstly invented **assembly language** and **subroutine library**
 - Triggered invention of **high-level languages**
 - Stored and built a collection of short library programs by **von Neumann** on punch cards
- Introduced the concept of **microprogramming** in 1951
 - Revolutionized CPU development efforts



John von Neumann (1903~1957)

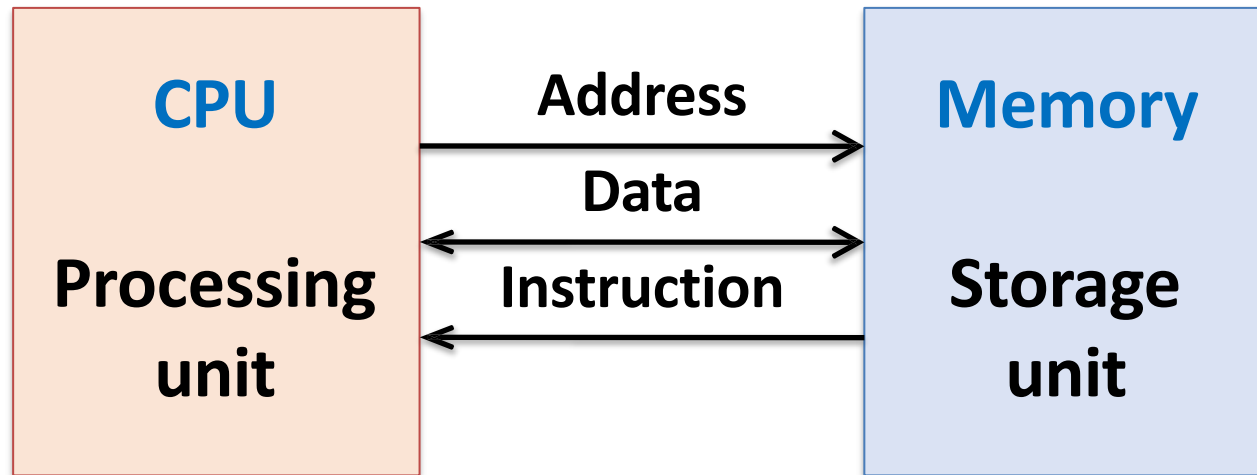
- Proposed the concept of modern computers in his paper "First Draft of a Report on the EDVAC" in 1945
 - The **concept of modern computers**
 - Idea of **storing programs internally** and **executing sequentially**
<Computers with memory and encoded operation orders that retrieve **stored information sequentially** to interpret instructions and execute operations>
 - Applied to computers such as EDSAC (1949) and EDVAC (1950)



John von Neumann (1903~1957) – Cont'd

- Institute for advanced study (ISA, 1952) computer
 - The name of a research institute of Princeton
 - The final completion of von Neumann architecture
 - Characteristics of IAS computer
 - Consists of main memory, arithmetic logic unit, program control unit, and input/output devices
 - Programs are executed in the order they are stored in memory
 - Types of commands including transfer, branch, arithmetic, and logic
 - Various registers are included in the control unit and arithmetic unit
 - Repeatedly performs instruction fetching and execution cycles

John von Neumann (1903~1957) – Cont'd



Data movement
Arithmetic & logical ops
Control transfer

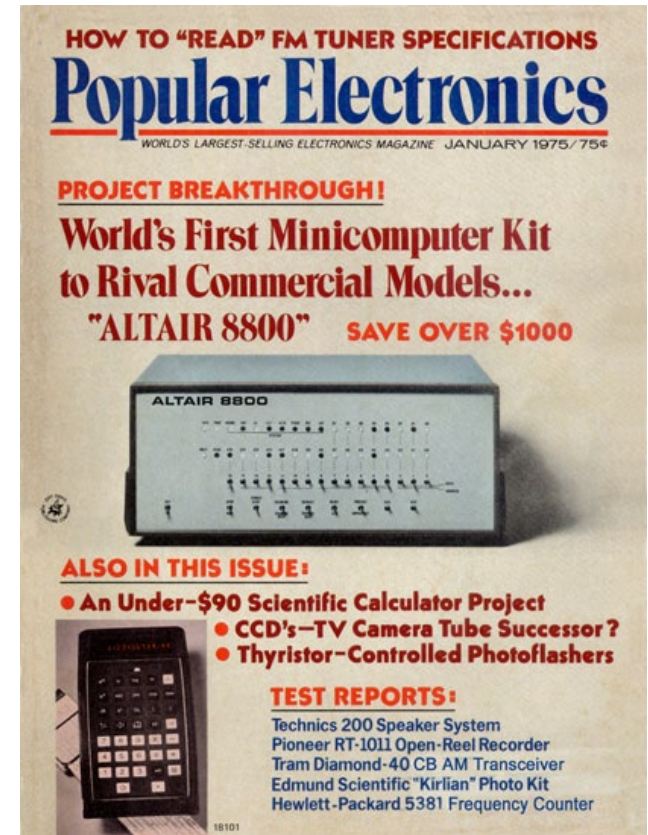
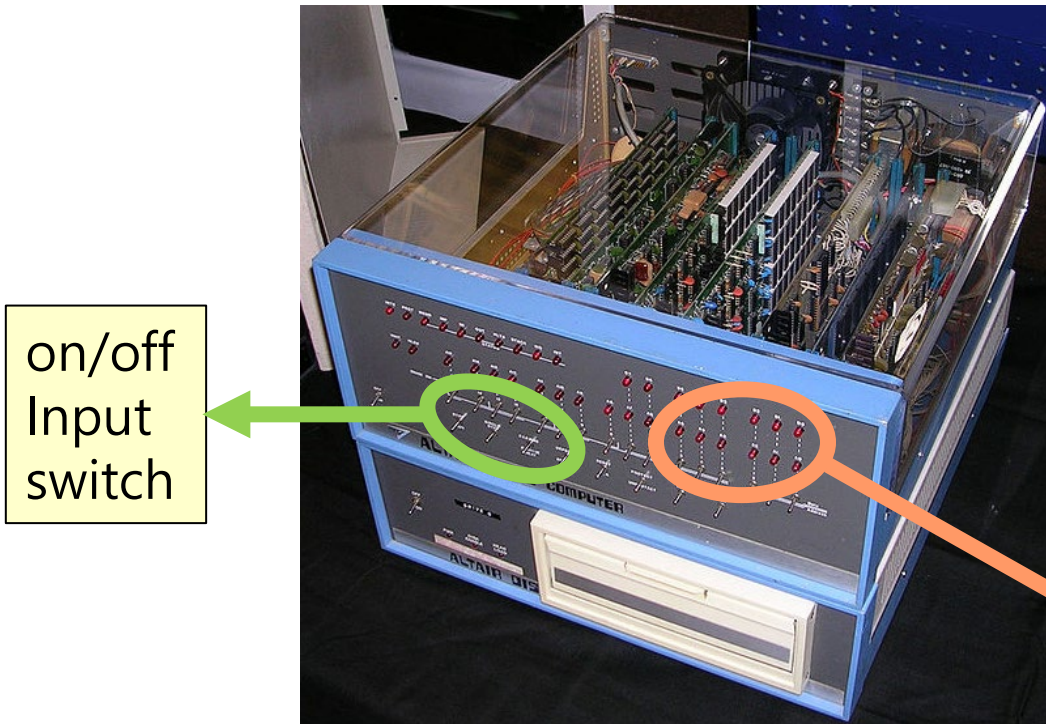
Byte addressable array
Code + data (user program, OS)
Stack to support procedures

Personal computer in 1970-80s

Year	Model	Detail	Bit
1975	Altair8800	The first commercial PC. Needs to assemble parts	8
1976	Apple I	Invented by Steve jobs and Wozniak	8
1977	Apple II	Popularized PC	8
1981	IBM PC	The original IBM PC	8
1983	IBM PC XT	8 bit PC by IBM	8
1984	IBM PC AT	16 bit PC by IBM	16
1984	Macintosh	Apple PC provided GUI	16, 32
1987	IBM PS/2	The first PS/2 series PC by IBM	16, 32
		...	

Altair 8800

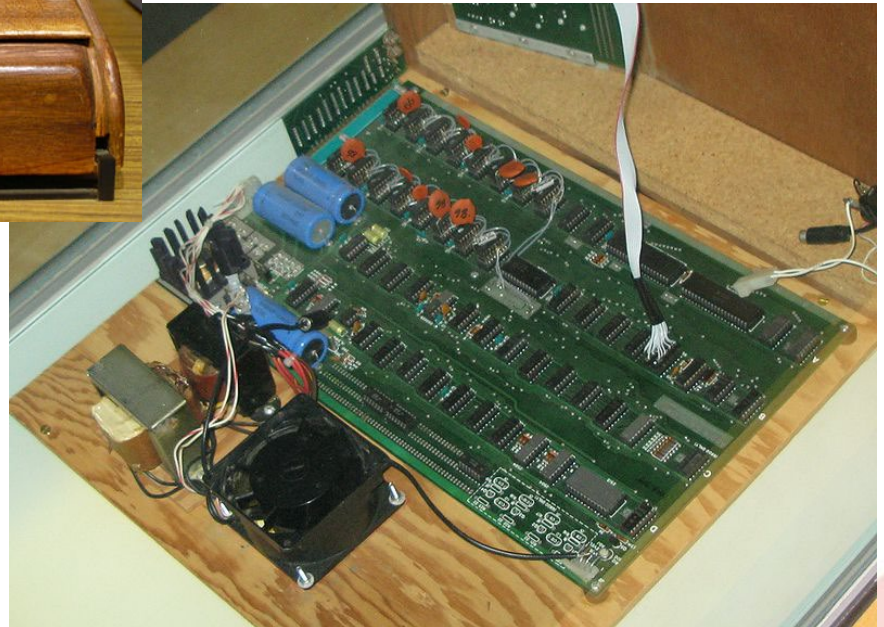
- Altair 8800 of MITS
 - Has no keyboard and mouse
 - Used **toggle switch** to input instructions
 - Results were indicated using **LED**



Apple I (1976)



User assembles the case, power supply, keyboard and video device



Apple II (1977)



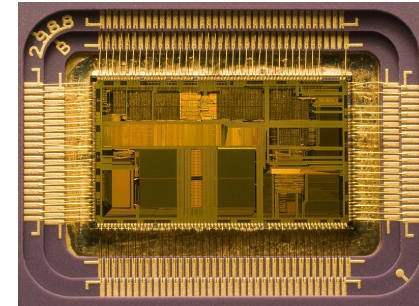
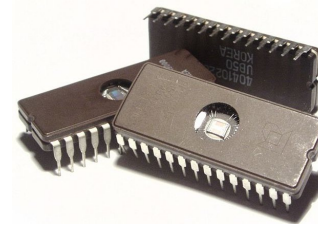
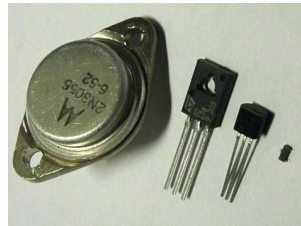
Provide external slots
and user-friendly OS

Attached
monitor and
FDD onto the
main body

Consists of
keyboard, sound,
8 external slots
and plastic case

Developments of electronic components

- Vacuum tubes → transistors → Integrated circuits (IC) → Microprocessor



- The integration density of transistors in integrated circuits

Category	Number of transistors
Small scale integration (SSI)	~100
Medium scale integration (MSI)	~1,000
Large scale integration (LSI)	~10,000
Very large scale integration (VLSI)	~1,000,000
Ultra large scale integration (ULSI)	1,000,000 ~

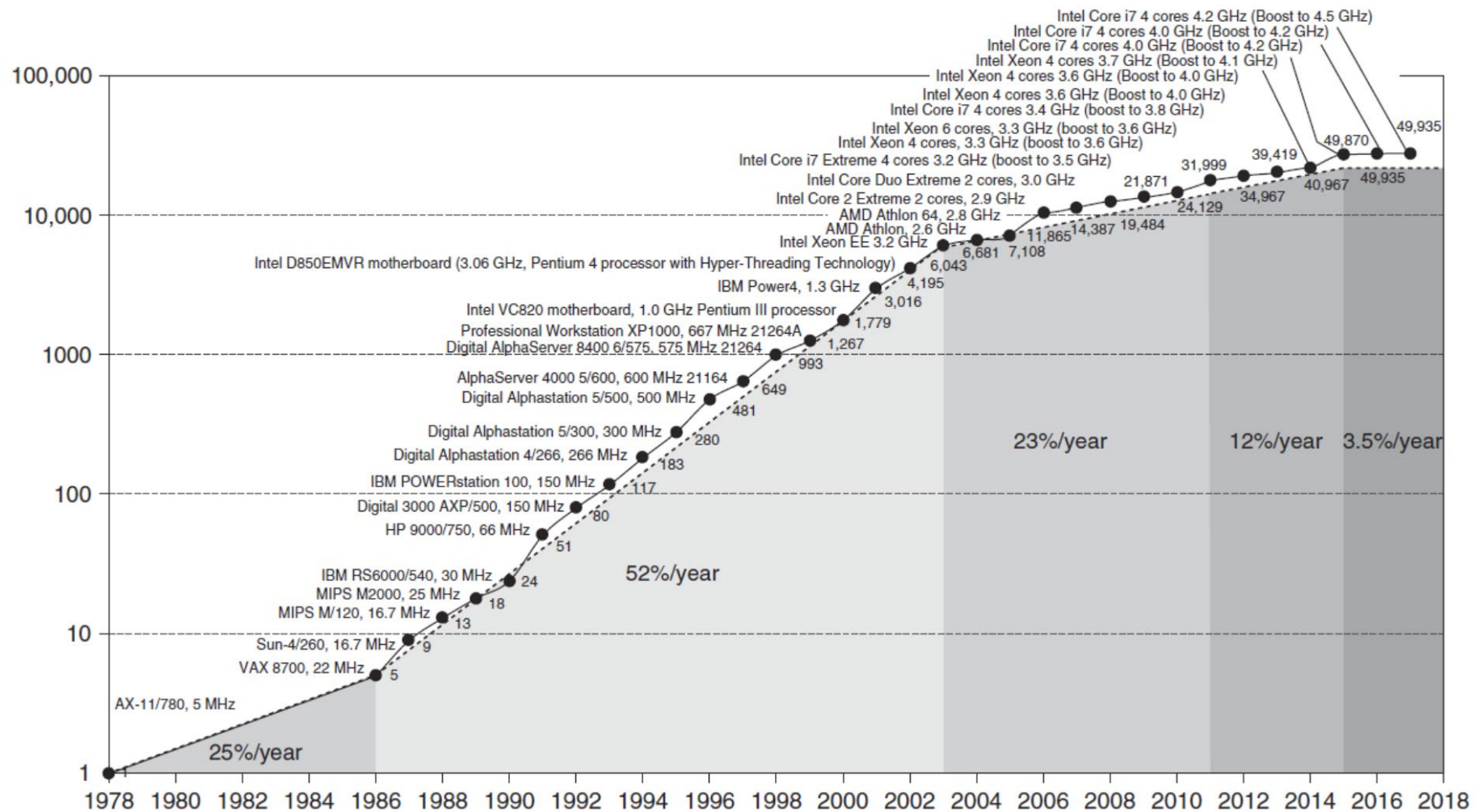
Generation of computers

- The integration density of transistors in integrated circuits

Category	Number of transistors
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Large scale integration (LSI)	~10,000
Very large scale integration (VLSI)	~1,000,000
Ultra large scale integration (ULSI)	1,000,000 ~

Category	Years	Electronic components
1 st generation	1940s	Vacuum tubes
2 nd generation	1950s	Transistors
3 rd generation	1960s	IC (SSI, MSI)
4 th generation	1970s	Microprocessor (LSI, VLSI, ULSI)
5 th generation	1980s and onwards	Microprocessor (VLSI, ULSI)

Growth in processor performance



Modern computer systems

Summary of the main stream computing classes

Feature	Personal mobile device (PMD)	Desktop	Server	Clusters/warehouse-scale computer	Internet of things/ embedded
Price of system	\$100–\$1000	\$300–\$2500	\$5000–\$10,000,000	\$100,000–\$200,000,000	\$10–\$100,000
Price of microprocessor	\$10–\$100	\$50–\$500	\$200–\$2000	\$50–\$250	\$0.01–\$100
Critical system design issues	Cost, energy, media performance, responsiveness	Price-performance, energy, graphics performance	Throughput, availability, scalability, energy	Price-performance, throughput, energy proportionality	Price, energy, application-specific performance

Summary of the main stream computing classes

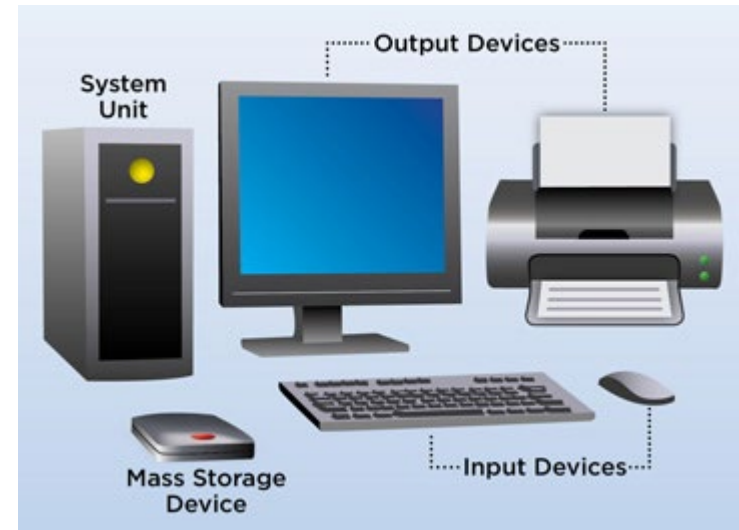


Google TV™

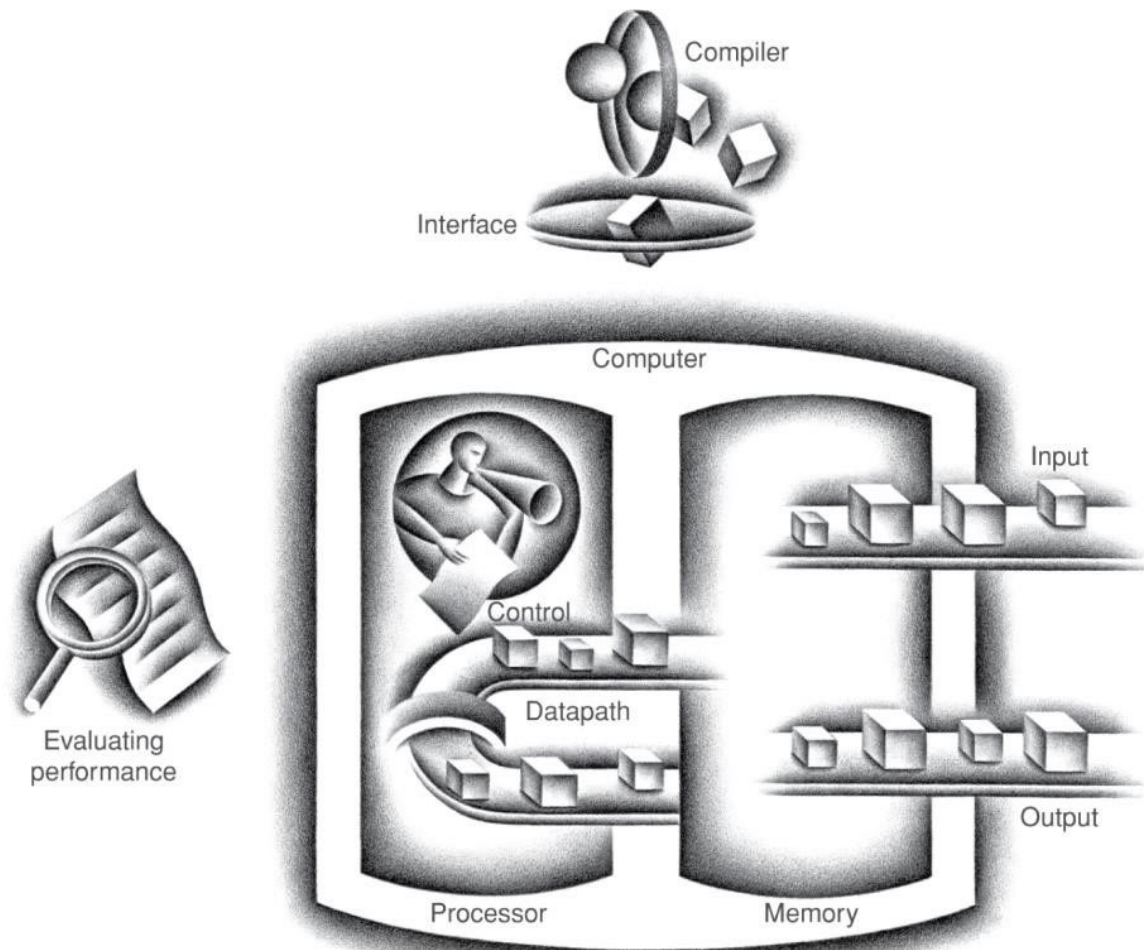


Components of PC

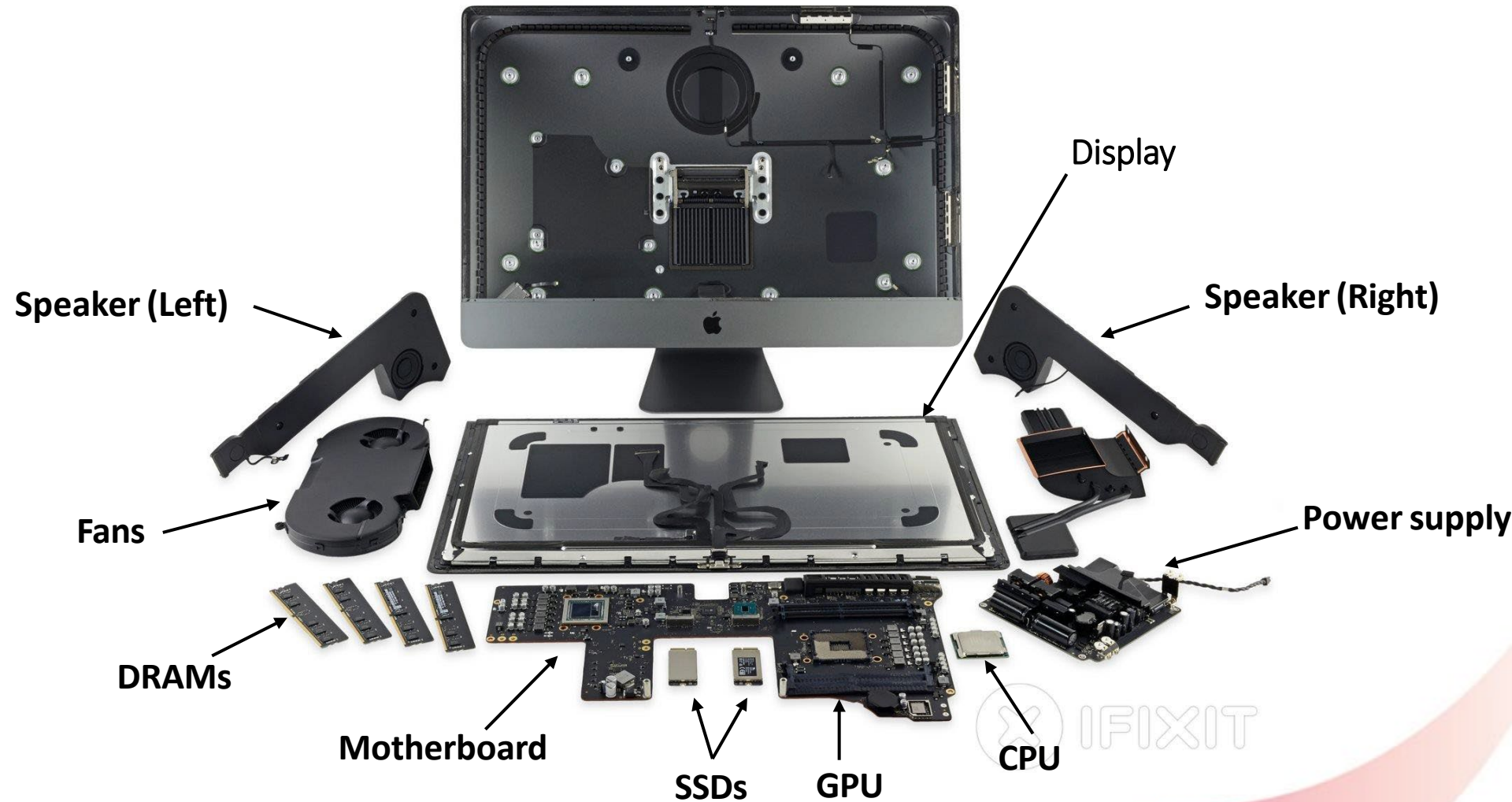
- CPU: Control + Datapath
- Memory
- I/Os
 - User-interface devices:
Display, keyboard, mouse, sound, ...
 - Storage devices:
HDD, SSD, CD/DVD, ...
 - Network adapters:
Ethernet, 3G/4G/5G, WiFi, Bluetooth, ...
- Same components for all kinds of computer



Components of PC – Cont'd

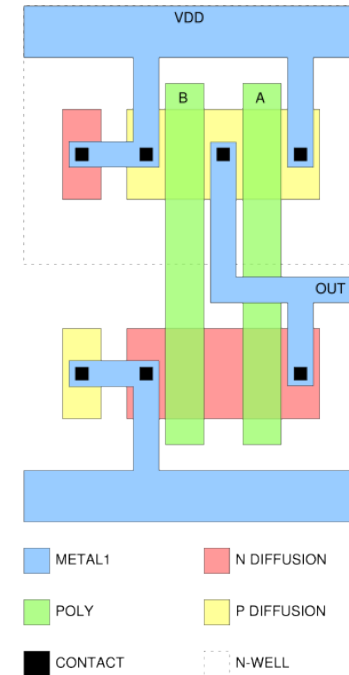
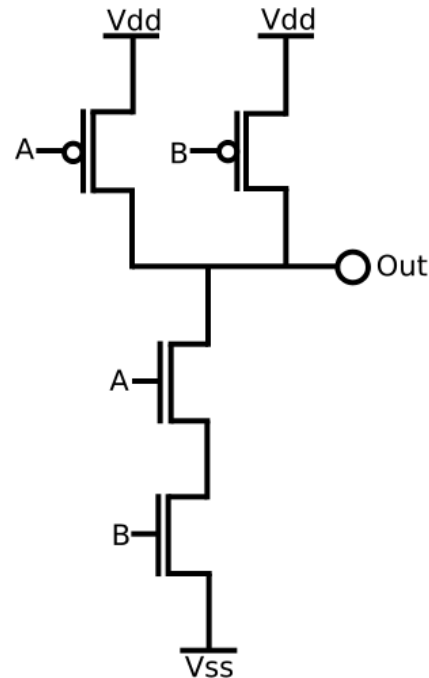


Modern PC (iMac Pro)



Transistors and logic gates

- NAND logic built with CMOS technology



Thank you!